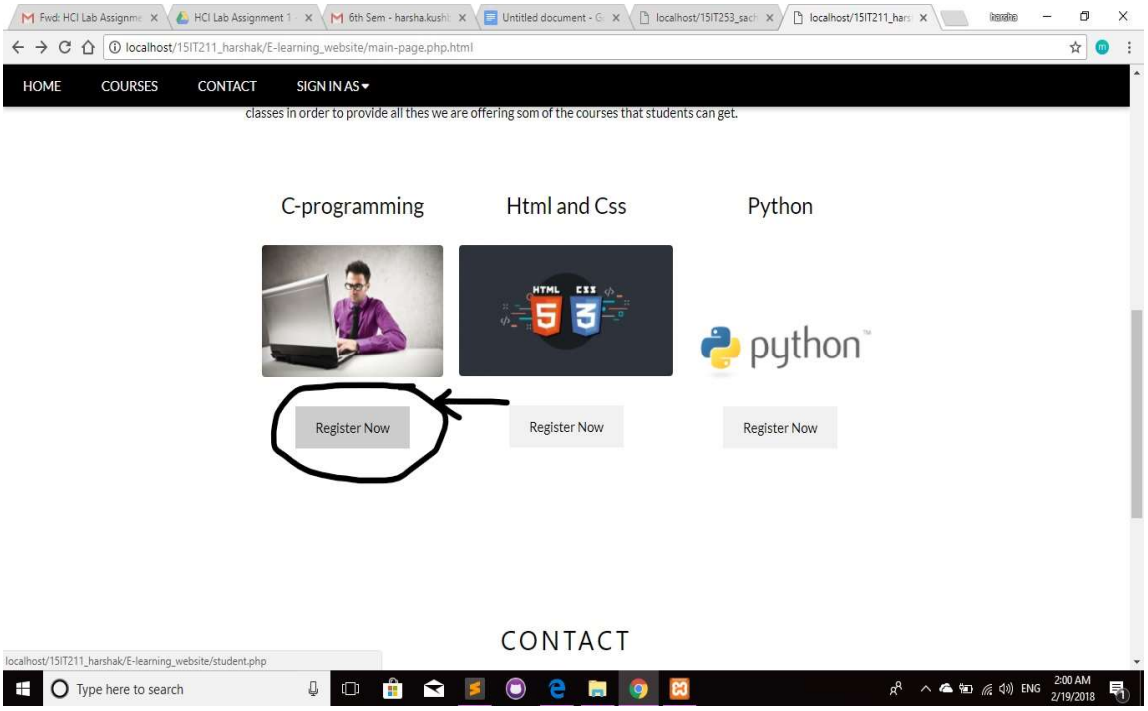
Design Rules

Shneiderman's Eight Golden Rules of Interface Design

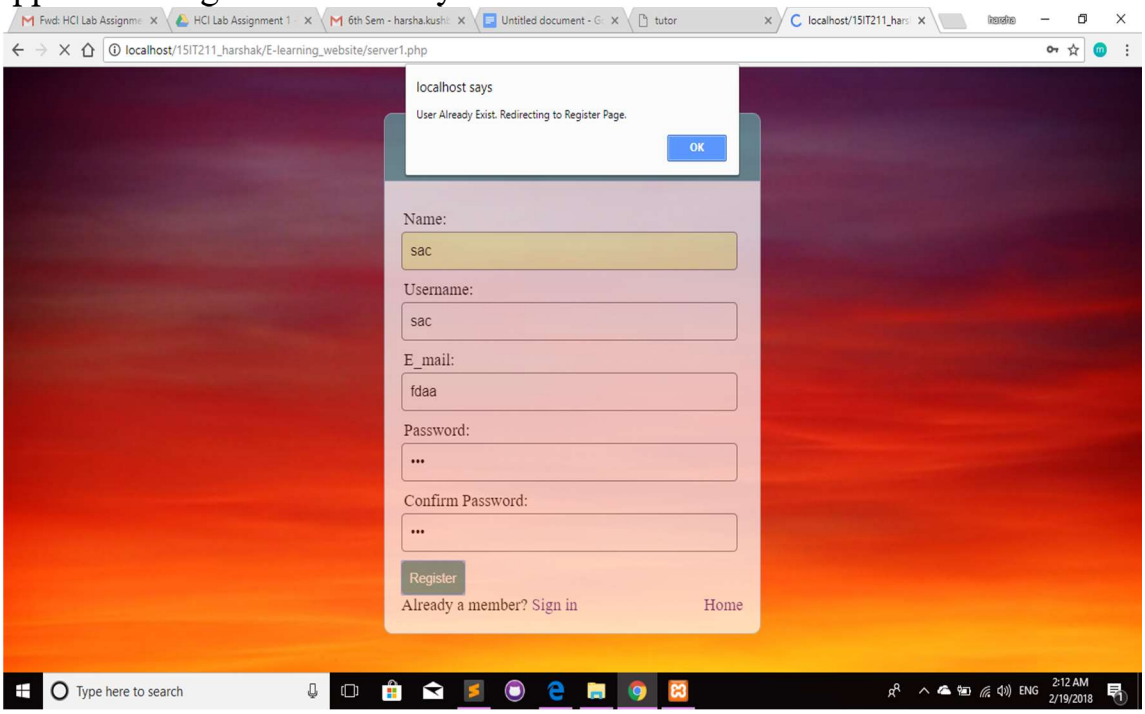
1. **Strive for consistency in action sequences, layout, terminology, command use and so on.**
   Consistency in action sequences, layout, terminology, command use etc.., has seen maintained.

2. **Enable frequent users to use shortcuts, such as abbreviations, special key sequences and macros, to perform regular, familiar actions more quickly.**
   Shortcuts such as Home and Register (at coursers offered section) buttons are provided so that user can go to home very quickly and Register button redirects user to Sign Up page, without this button user would have required to follow Sing In As student → Sign Up.



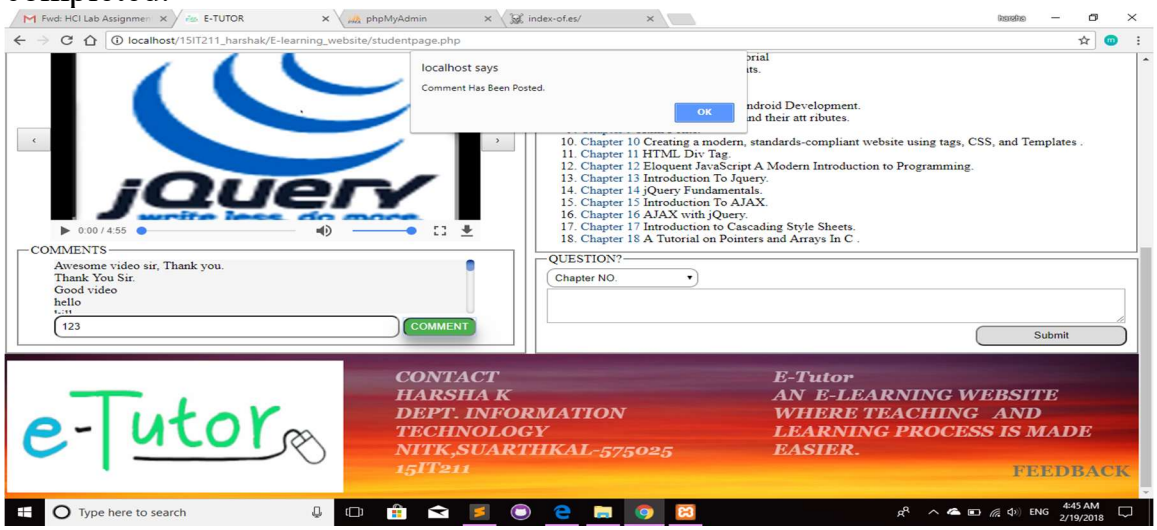
3. **Offer informative feedback for every user action, at a level appropriate to the magnitude of the action.**

When a new user tries to Sign up with existing user id an informative feedback appears stating an user already exist.

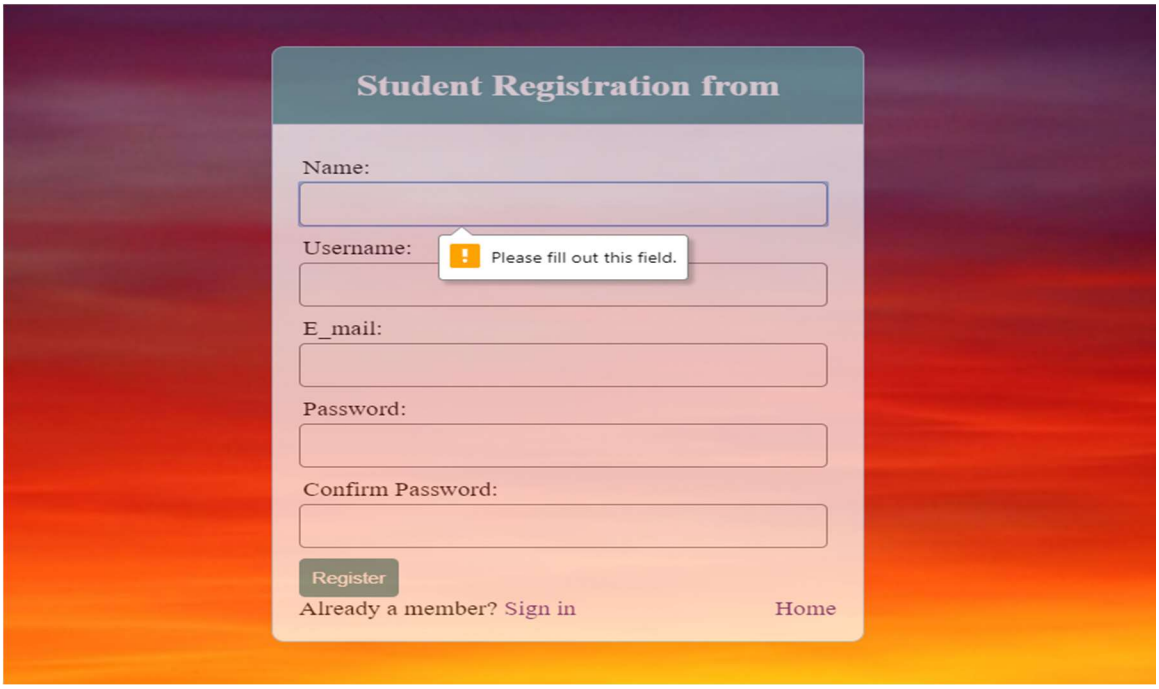


4. **Design dialogs to yield closure so that the user knows when they have completed a task.**

When user submits the comment or feedback form or asks a question a dialog will appear which yields closure so that user comes to know that his/her task of submitting comment or feedback form or a question has been successfully completed.



5. **Offer error prevention and simple error handling so that, ideally, users are prevented from making mistakes and, if they do, they are offered clear and informative instructions to enable them to recover.**
   When user press register button without filling all the required fields website offers error prevention by not letting user to perform Sing Up action and pop ups a message to fill required field. And all other error are anticipated and system offers informative instructions to enable user to recover them.



6. **Permit easy reversal of actions in order to relieve anxiety and encourage exploration, since the user knows that he can always return to the previous state.**
   Every page contains reversal button that lets one to go back to the previous state.

7. **Support internal locus of control so that the user is in control of the system, which responds to his actions.**
   The website is made such that user always has control of it, and each element of the webpage responds correctly to the user action.

8. Reduce short-term memory load by keeping displays simple, consolidating multiple page displays and providing time for learning action sequences.

The webpages in the E-learning system have simple displays and pages that contain video tutorials and notes have been combined together and made to be in single page.

Norman's Seven Principles

1. **Use both knowledge in the world and knowledge in the head. People work better when the knowledge they need to do a task is available externally – either explicitly or through the constraints imposed by the environment.**
   Knowledge in the world and own knowledge has been applied while creating the E-learning system. World knowledge has been used to avail the features that one expect it to be present and own knowledge is used avail addition features to the user.

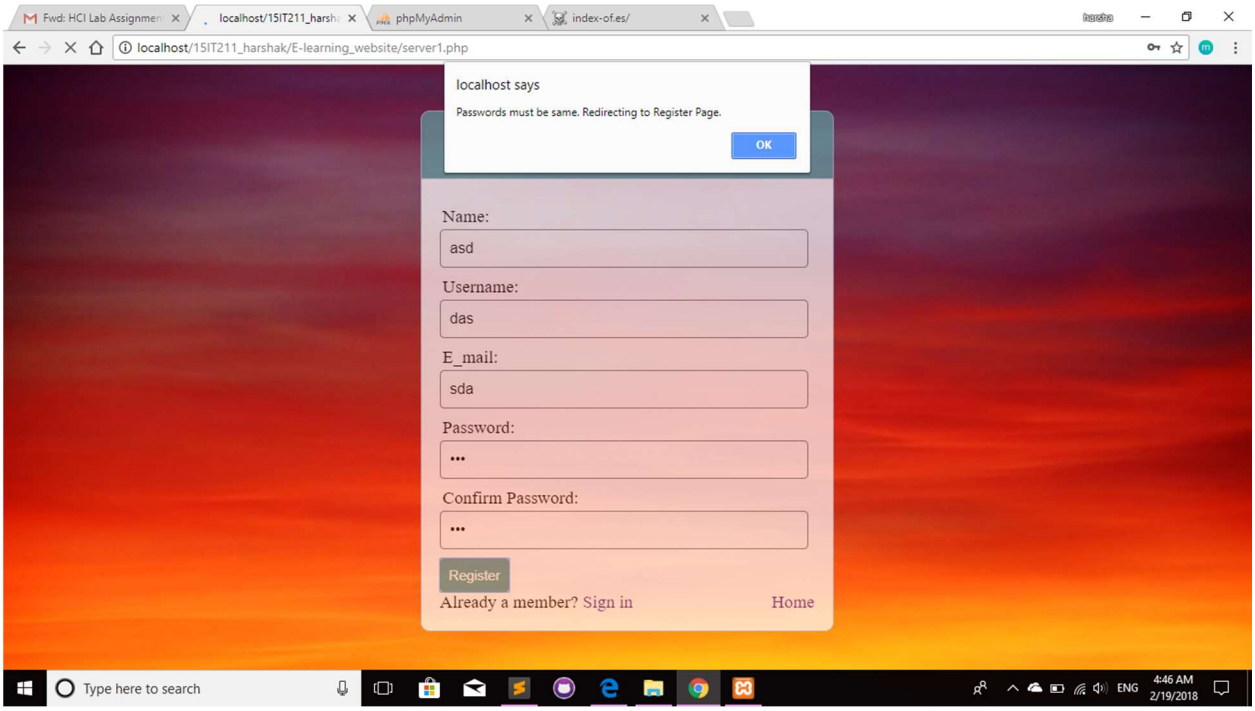
2. **Simplify the structure of tasks. Tasks need to be simple in order to avoid complex problem solving and excessive memory load.**
   Usage of available technology has provided the user with more information about the task and better feedback. Tasks like comment updation, feedback submitted till now and questions asked till now are automated so that user know the information that system possess. All the tasks in the system are simple in nature and none of them are intended to the control away from the user.

3. **Make things visible: bridge the gulfs of execution and evaluation. The interface should make clear what the system can do and how this is achieved, and should enable the user to see clearly the effect of their actions on the system.**
   All the changes that task and user action make to system are clearly visible within the system. When user comments, submit a feedback and ask a question the changes that result into the system can be seen clearly.

4. **Get the mappings right. User intentions should map clearly onto system controls. User actions should map clearly onto system events. So it should be clear what does what and by how much.**
   User intentions are mapped clearly onto system controls. The system is in full control of user and his/her actions. User action are mapped to the system events that is whenever user action is performed the event corresponding to the respective action is triggered within the system.

5. **Exploit the power of constraints, both natural and artificial. Constraints are things in the world that make it impossible to do anything but the correct action in the correct way.**
   All the actions are ensured to be done in the correct way. None of the action in the system produce incorrect result. All the constraints are looked after to ensure correct working of the system.

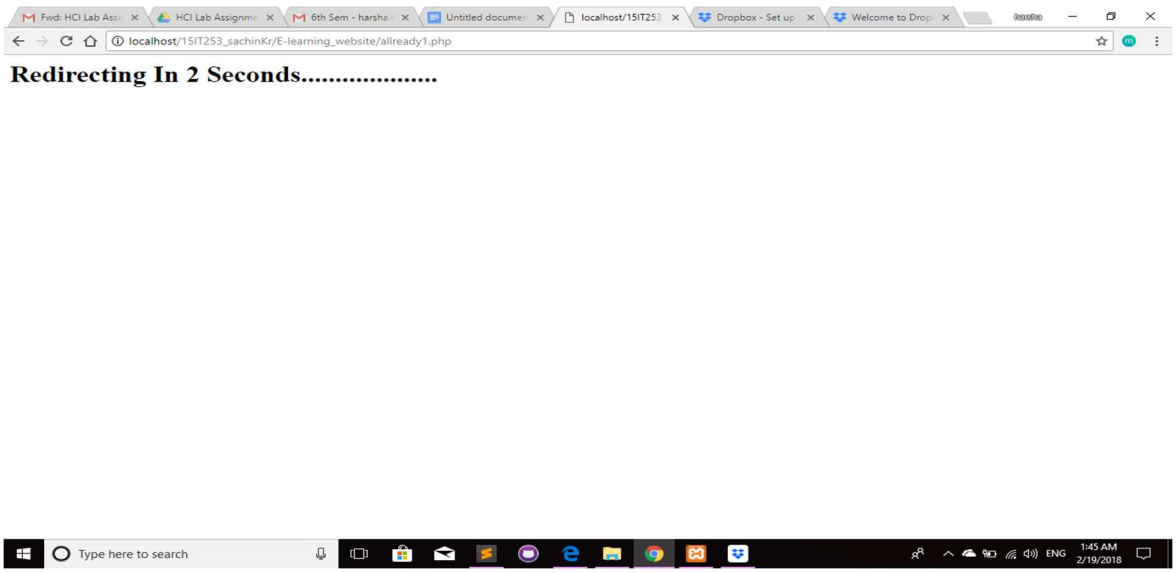
6. **Design for error. To err is human, so anticipate the errors the user could make and design recovery into the system.**
   Error like non-existing user trying to login, providing two different passwords while registering and not filling the required field while registering have been anticipated and the system produces corrective pop up messages in case user performs these error.



7. **When all else fails, standardize. If there are no natural mappings then arbitrary mappings should be standardized so that users only have to learn them once.**
   It has been seen that none of the mappings are failed therefore there is no need of making use of arbitrary mappings in order to standardization.

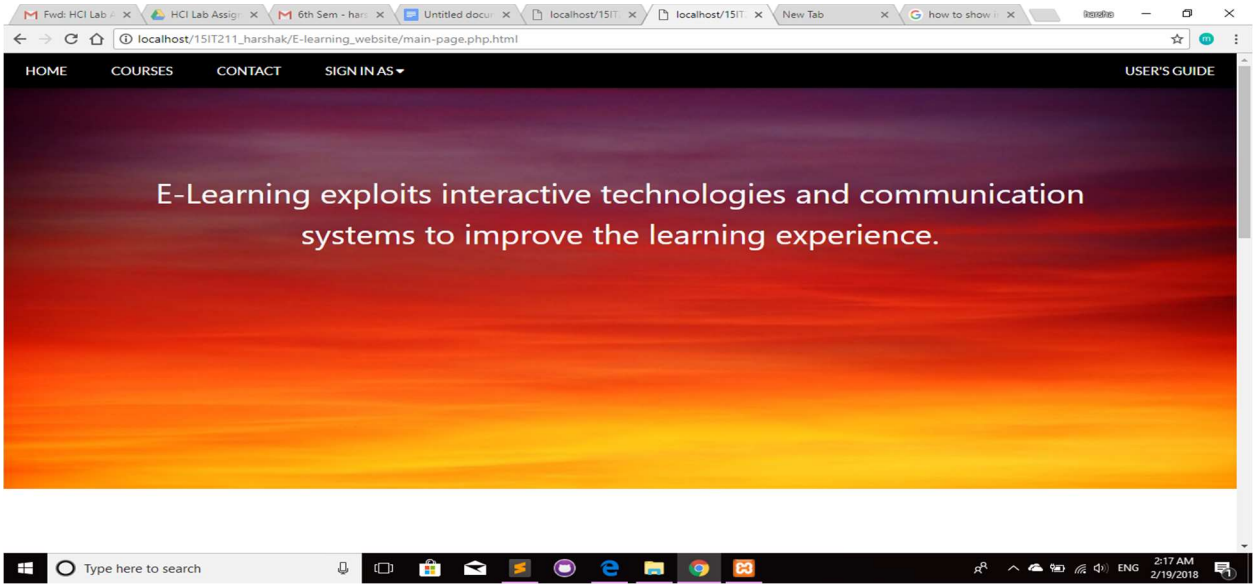
Nielsen’s ten heuristics

1. **Visibility of system status : Always keep users informed about what is going on, through appropriate feedback within reasonable time.**



2. **Match between system and the real world :The system should speak the user’s language, with words, phrases and concepts familiar to the user, rather than system-oriented terms.**

System speak English language as it is one of the most used language in the world.



3. **User control and freedom : Users often choose system functions by mistake and need a clearly marked 'emergency exit' to leave the unwanted state without having to go through an extended dialog. Support undo and redo.**

"Sign Out" button has been provided in case if user wants to leave the system.

4. **Consistency and standards : Users should not have to wonder whether words, situations or actions mean the same thing in different contexts.**

None of the actions cause two or more result. Each action produce single result and have same meaning in every situation.

5. **Error prevention : Make it difficult to make errors. Even better than good error messages is a careful design that prevents a problem from occurring in the first place.**

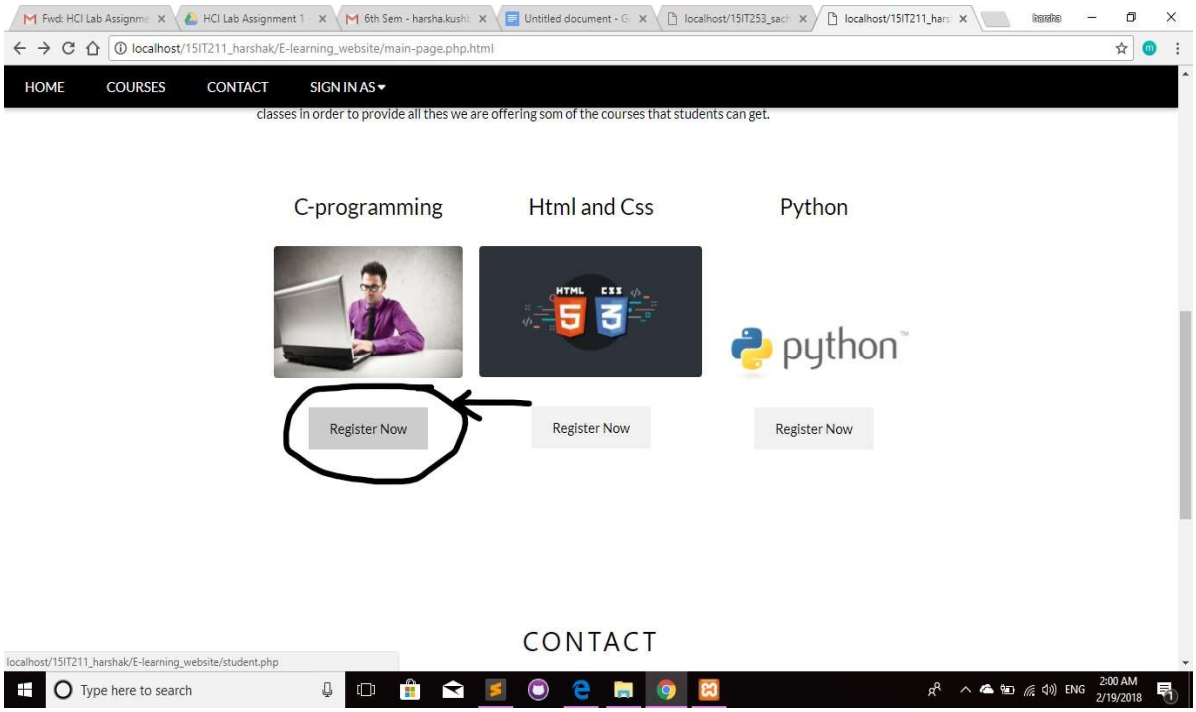
Error like non-existing user trying to login, providing two different passwords while registering and not filling the required field while registering have been anticipated and the system produces corrective pop up messages in case user performs these error.

6. **Recognition rather than recall : Make objects, actions and options visible. The user should not have to remember information from one part of the dialog to another.**

System is designed such a way that user remembers the information easily. The objects, actions and options are visible clearly in the website.

7. **Flexibility and efficiency of use : Allow users to tailor frequent actions.**

Shortcuts such as Home and Register (at coursers offered section) buttons are provided so that user can go to home very quickly and Register button redirects user to Sign Up page, without this button user would have required to follow Sing In As student → Sign Up.

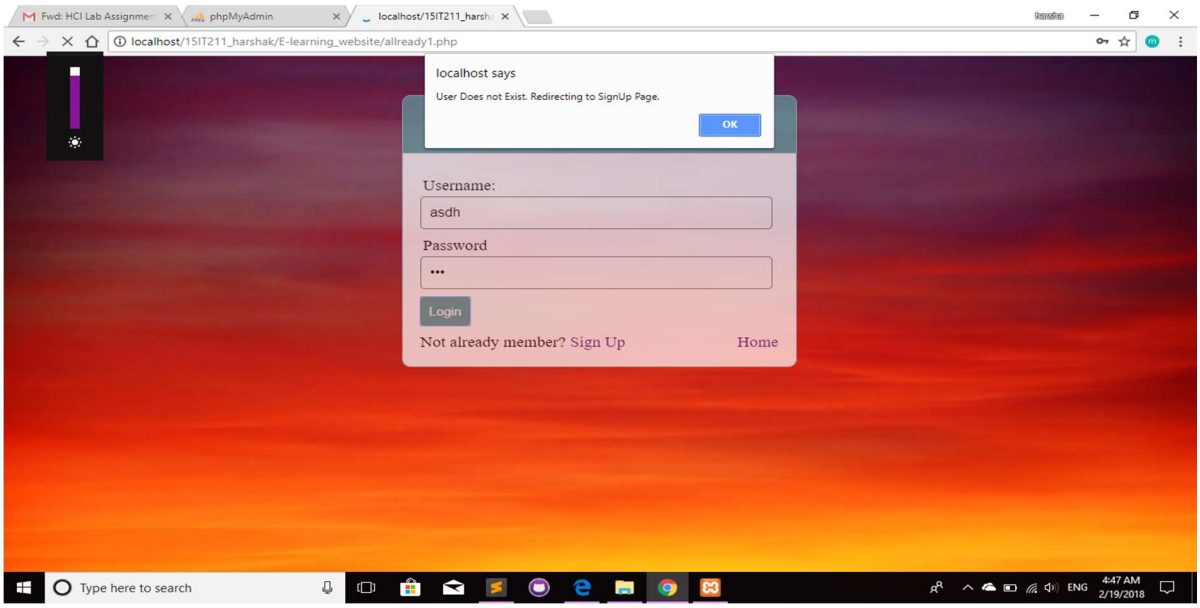


8. **Aesthetic and minimalist : Design Dialogs should not contain information that is irrelevant or rarely needed. Every extra unit of information in a dialog competes with the relevant units of information and diminishes their relative visibility.**

Dialog box are designed such that they contain the information that is relevant and needed. None of the Dialog box is designed with irrelevant information or extra unit of information that in not needed.

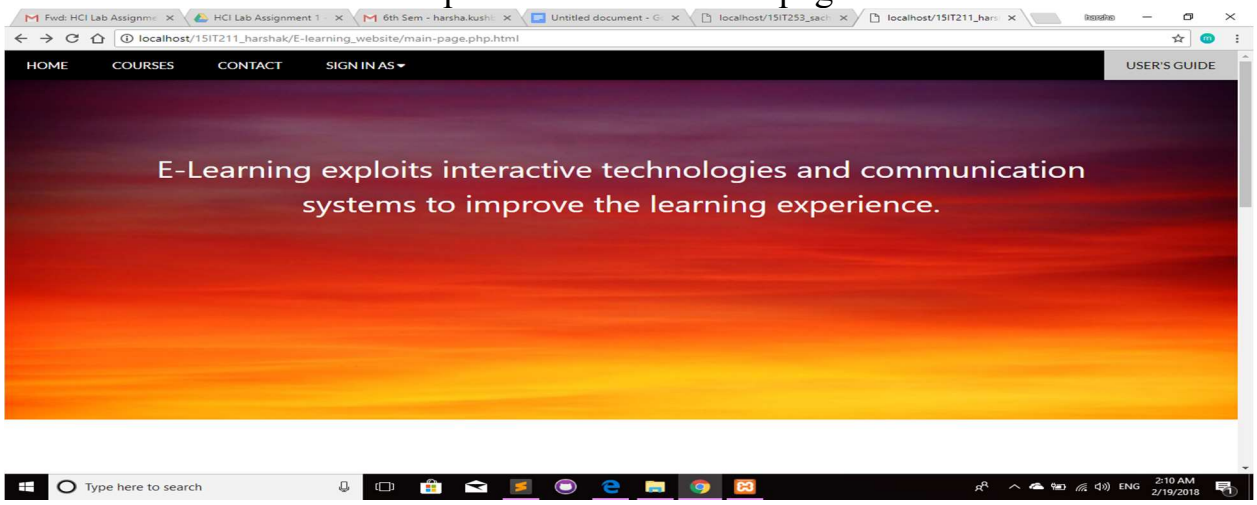
9. Help users recognize, diagnose and recover from errors : Error messages should be expressed in plain language, precisely indicate the problem, and constructively suggest a solution.

System is built such a way that it helps user recognize, diagnose and recover from error. Error message has been expressed in plain language, and indicate the exact problem that exists.



10. Help and documentation Few systems can be used with no instructions so it may be necessary to provide help and documentation.

User document has been provided in the Home page.



UNIVERSAL DESIGN

Module 1 :Universal Design

1.Principle one is equitable use**:** The design is useful to people with a range of abilities and appealing to all.We have used this between both tutor and student user.

2.Principle two is flexibility in use: The design allows for a range of ability and preference, through choice of methods of use and adaptivity to the user's pace, precision and custom.

This system can be used in any computer provided having good internet connectivity.

3.Principle three is that the system be simple and intuitive to use: Regardless of the knowledge, experience, language or level of concentration of the user.The system is involving simple features and very attractive features which make the users to use more intuitively.

4.Principle four is perceptible information:The design should provide effective communication of information regardless of the environmental conditions or the user's abilities. Redundancy of presentation is important: information should be represented in different forms or modes (e.g. graphic, verbal, text, touch.The information provided in the system is perceptible.

5.Principle five is tolerance for error: Minimizing the impact and damage caused by mistakes or unintended behavior.The system has warning before doing mistake by users and its recovery is done easily.

6.Principle six is low physical effort: systems should be designed to be comfortable to use, minimizing physical effort and fatigue.The buttons are kept in right place for users to return back to their desired place.

7.Principle seven requires size and space for approach and use: the placement of the system should be such that it can be reached and used by any user regardless of body size, posture or mobility. The system uses proper size of images to upload as assignments.

Module 2 :Designing for diversity

This system is designed by keeping every age group people.It's useful for every users who is using it.

COMMUNICATION and COLLABORATION MODELS

Module 1: Face-to-face communication:

Back channels, confirmation and interruption

We have confirmation and interruption if users enters wrong password and gives confirmation on login by showing his name.

Module 2: Text-based communication

For asynchronous groupware (and even some synchronous systems), the major form of direct communication is text based.We have used Text based communication for feedback and comments from users.

MODELLING RICH INTERACTION

Module 1: Status–Event Analysis

Mouse When the user is positioning the mouse pointer over a target, his attention will be focussed on that target. This is guaranteed in all but a few situations by the feedback requirements of hand–eye coordination. However, this attention may not stay long after the target has been successfully 'hit'.We are using this analysis by keeping this feature for certain things like going back to homepage through highlighting that part of button for clicking.

Module 2: Low Intention and Sensor-Based Interaction

In fact, there is a continuum of intentionality . At one extreme are 'normal' intentional interactions such as pressing the computer key or not pressing the key pressing the submit button in our case.

HYPERTEXT, MULTIMEDIA and WWW

Module 1: Static Web Content

Static content is published to regular files on your server and handled using the simplest methods available to the web server. The advantages of static content are:

- it is the fastest and most efficient way to deliver content.We have this by sending variables to each PHP page.
- it uses simple, clean URLs to address the content.We have used clear website and certain urls for loading images.

Module 2: Dynamic Web Content

Dynamic content is generated for you at the time you request the page. The document you view exists only for you at that moment; if viewed by someone else at the same time, or by you at a slightly different time, you could get something different. Dynamic content is good for:

- pages whose content changes too quickly to easily Re-publish it.Our pages changes quickly from one page to another page without time delay.
- pages that display viewer-specific content (eg. user profiles).We have seperate profile for seperate users like student.
- pages that display content conditionally (ie. member-only pages).We display only photos of certain users on login of their profile.